

Limits

(1)

$$\lim_{x \rightarrow \frac{\pi}{6}} \frac{2\sin x - 1}{2\cos x - \sqrt{3}}$$

$$\Rightarrow \sin\left(\frac{\pi}{6}\right) = \frac{1}{2} \Rightarrow \frac{2\frac{1}{2} - 1}{2\frac{\sqrt{3}}{2} - \sqrt{3}} = \frac{0}{0}$$
$$\Rightarrow \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$$

L'Hospital Rule:

$$\frac{d}{dx}(2\sin x - 1) = 2\cos x$$

$$\frac{d}{dx}(2\cos x - \sqrt{3}) = 2\sin x - \frac{0}{2\sqrt{3}} = -2\sin x$$

So:

$$\frac{2\cos x}{-2\sin x}$$

Then:

$$\frac{\frac{2\sqrt{3}}{2}}{\frac{-2}{2}} = -\sqrt{3}$$

$$\lim_{x \rightarrow \frac{\pi}{6}} \frac{2\sin x - 1}{2\cos x - \sqrt{3}} = \sqrt{3}$$



(2)

$$\lim_{x \rightarrow \frac{\pi}{3}} \frac{1 - 2\cos x}{\sin x - \frac{\sqrt{3}}{2}}$$

$$\Rightarrow \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

$$\Rightarrow \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \frac{1 - 2 \cdot \frac{1}{2}}{\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}} = \frac{0}{0}$$

L'Hôpital Rule:

$$\frac{d}{dx}(1 - 2\cos x) = +2\sin x$$

$$\frac{d}{dx}\left(\sin x - \frac{\sqrt{3}}{2}\right) = \cos x - \frac{0}{0} = \cos x$$

$$\text{So: } \Rightarrow \frac{2\sin x}{\cos x}$$

$$\text{Then: } \frac{\frac{2\sqrt{3}}{2}}{\frac{1}{2}} = 2\sqrt{3}$$

$$\lim_{x \rightarrow \frac{\pi}{3}} \frac{1 - 2\cos x}{\sin x - \frac{\sqrt{3}}{2}} = 2\sqrt{3}$$



(3)

$$\lim_{x \rightarrow 0} \frac{\sin(3x) - 3x}{x^3}$$

$$\Rightarrow \frac{\sin(0) - 3(0)}{0^3} = \frac{0}{0} \quad \text{I.F.}$$

L'Hôpital Rule:

$$\frac{d}{dx}(\sin(3x) - 3x) = 3\cos(3x) - 3$$

$$\frac{d}{dx}(x^3) = 3x^2$$

$$\text{So: } \frac{3\cos(3x) - 3}{3x^2}$$

$$\text{Then: } \frac{3\cos(0) - 3}{3(0)^2} = \frac{0}{0}$$

$$\frac{d}{dx}(3\cos(3x) - 3) = (9(-\sin(3x)))$$

$$\frac{d}{dx}(3x^2) = 6x$$

$$\Rightarrow \frac{(9)(-\sin(3x))}{6x} = \frac{0}{0}$$

$$\frac{d}{dx}(-9\sin(3x)) = -9\cos(3x) \cdot 3 = -27\cos(3x)$$

$$\frac{d}{dx}(6x) = 6$$

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$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$$

$$\Rightarrow \frac{e^{2(0)} - 1 - 2(0)}{(0)^2} = \frac{1 - 1}{0} = \frac{0}{0} \quad \text{I.S.}$$

L'Hôpital Rule:

$$(e^{2x} - 1 - 2x)' = 2e^{2x} - 2$$

$$(x^2)' = 2x$$

$$\Rightarrow \frac{2e^{2x} - 2}{2x} = \frac{0}{0}$$

$$(2e^{2x} - 2)' = 4e^{2x}$$

$$(2x)' = 2$$

$$\Rightarrow \frac{4e^{2x}}{2} = \frac{4(1)}{2} = 2$$

$$\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2} = 2$$

